Fiscal Year:	FY 2022	**Task Last Updated:**	FY 01/06/2024
PI Name:	Haggett, Lindsey Ph.D.		
Project Title:	Mechanisms of Protection from Radiation by Natural Proteins		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	haggett@bcm.tmc.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	713-798-8730
Organization Name:	Baylor College of Medicine, Inc		
PI Address 1:	1 Baylor Plz Rm S809A MailStop BCM#225		
PI Address 2:			
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77030-3411	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 TRI-RFA-1802-PD: Translational Research Institute for Space Health (TRISH) Postdoctoral Fellowships
Start Date:	12/01/2018	**End Date:**	11/30/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 11/30/2021 per TRISH (Ed., 10/1/20)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rosenberg, Susan Ph.D. (MENTOR: Baylor College of Medicine)		
Grant/Contract No.:	NNX16AO69A-P0302		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

POSTDOCTORAL FELLOWSHIP
Rationale: Individuals in high-radiation environments, including space, suffer DNA damage, which increases their susceptibility to cancer, among other diseases. DNA-damage levels in cells can be used to measure the extent to which cells succumb to, or alternatively resist, deleterious consequences of radiation. This project leverages proteins that confer lower-than-normal levels of spontaneous DNA damage to cells to discover molecular mechanisms of protection from radiation by overproduced natural proteins. The ultimate goal is to rationalize design of drugs to upregulate human proteins for protection from radiation in space.

Unique resource: The Rosenberg lab identified Escherichia coli proteins that alter levels of endogenous DNA damage when overproduced–208 that increase and 23 that decrease damage, the latter relevant to radiation resistance. The human-protein relatives of the bacterial “damage-up” proteins also increased DNA damage when overproduced in human cells, and are overrepresented among cancer-driving proteins. These data demonstrate the relevance and power of conserved bacterial genes for discovery of important human biology of DNA damage. Preliminary results from the parent grant indicate that overproduction of a subset of the bacterial “damage-down” proteins protects cells from exogenous radiation, conferring reduced DNA damage. However, the mechanisms of protection remain to be determined.

Plan: This project will dissect possible mechanisms of protection by the identified radiation-protection proteins (RPPs). We will (1) determine whether identified RPPs block production of DNA damage or increase DNA repair; (2) map genomic regions affected by RPP-overproduction; (3) determine whether mutagenesis is reduced by RPP-overproduction.

Deliverables and significance: Results of this fellowship will illuminate how RPPs reduce DNA damage levels, whether particular genomic regions (and which) are more susceptible to/protected from DNA-damage, and whether reducing DNA damage protects against disease-causing mutagenesis. These facets of the radiation-protection mechanisms will guide selection of related human proteins for overproduction for protection from radiation in space.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

OVERVIEW
Previously, the Rosenberg lab identified a set of 23 conserved bacterial proteins that, when overproduced, reduce levels of endogenous DNA damage in Escherichia coli cells. This project aims to build on that discovery by investigating the molecular mechanisms by which these DNA-damage-suppressing proteins (DDSPs) protect living cells, as well as identifying members that confer resistance to exogenous DNA damage from radiation. We will provide a list of human-relevant bacterial Radiation Protection Proteins (RPPs) and information about how they reduce DNA damage at the molecular level in order to streamline research into radiation countermeasures in human cells. The aims of our original proposal are as follows:

Aim 1: Determine whether damage-down genes prevent DNA damage or increase DNA repair.

Aim 2: Map genomic regions affected by damage-down overexpression.

Aim 3: Determine genome stabilities of damage-down cells.

KEY FINDINGS

1. We refined the conditions of our exogenous DNA damage assays to improve the detection of DNA damage levels, and we developed a strategy to verify the integrity of the reporter protein under conditions of RPP overproduction.

2. Overproduction of at least 13 of our 23 candidate proteins significantly reduces DNA damage in response to ionizing radiation.

3. Of the 13 proteins we have confirmed as RPPs, at least 9 had an endogenous/exogenous damage quotient of >1 when compared with the empty vector control strain, indicating that they decreased exogenous DNA damage levels further than was anticipated based on their lower level of endogenous DNA damage.

4. For 22 out of 23 candidate proteins, the rate of spontaneous mutagenesis is not significantly altered or is very modestly increased (< 2-fold) in cells overproducing RPPs.

5. We are currently measuring ROS production/clearance and monitoring cell survival after radiation exposure for cells overproducing our 13 confirmed RPPs, and we plan to complete additional assays to quantify DNA damage events, assess homologous recombination rates, and measure chromosome loss and protein stability in RPP-protected cells after the conclusion of the funding period.

IMPACT

1. Optimization of our experimental conditions and the additional controls we have developed will ensure proper detection of DNA damage reduction and rule out other possible explanations for the decrease in SOS signal observed during RPP overproduction.

2. Over 55% of the proteins we initially identified as reducing endogenous DNA damage are also capable of reducing exogenous DNA damage in live cells exposed to radiation. These proteins have homologs and functional analogs in human cells which perform similar functions, and those homologs and analogs may confer similar benefits to human cells when overproduced.

3. Genome instability is a significant driving factor in the development of diseases such as cancer, and none of our RPPs appear to jeopardize the genomic stability of bacterial cells.

RESEARCH PLAN

This was the final year of TRISH funding for this project.

Bibliography Type:	Description: (Last Updated: 01/06/2024)
Awards	Haggett L. "Dean's Award for Excellence," Dec-2018